

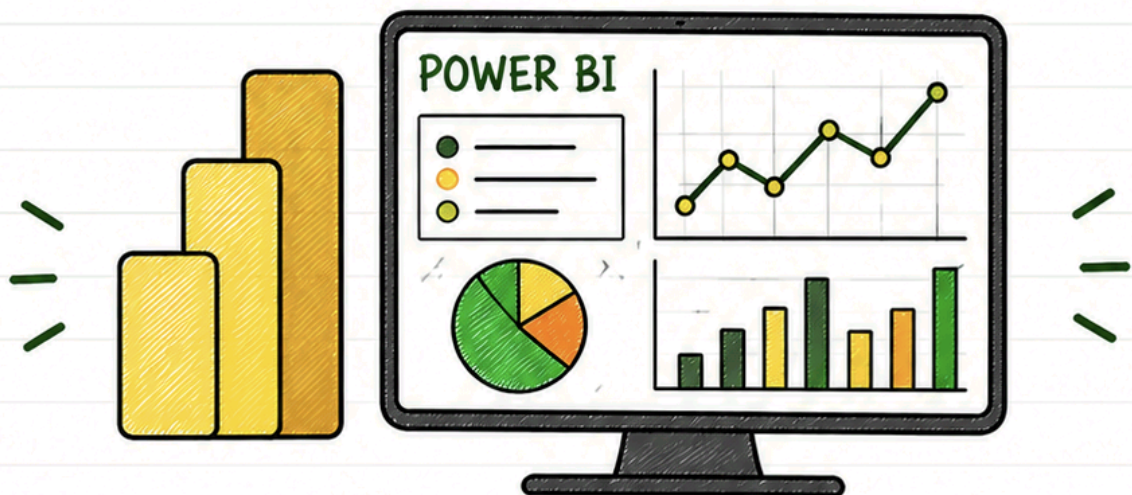


@qa_insights

POWER BI

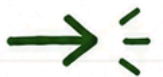
COMPLETE NOTES

(Power BI)



BEGINNER TO ADVANCED

SWIPE TO NEXT



FOLLOW



SHARE



COMMENT



LIKE

@qa_insights | Quality Insights, Better Testing!

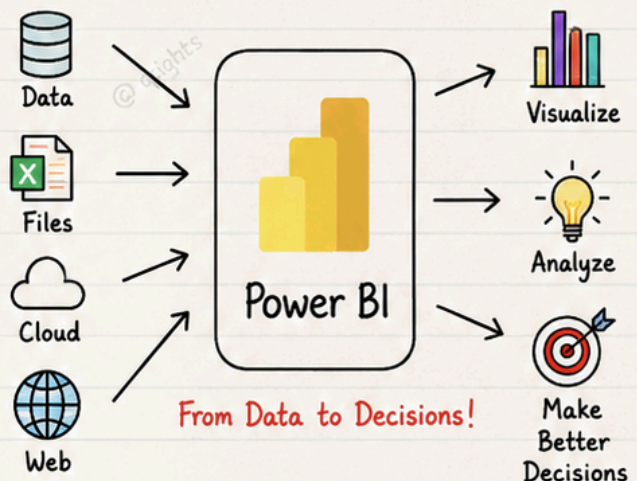
Introduction to Power BI

① What is Business Intelligence (BI)?

- BI is the process of collecting, analyzing and turning data into meaningful insights to support better business decisions.

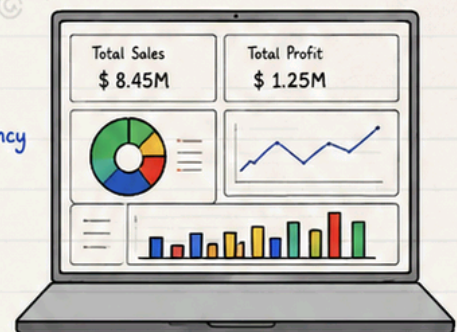
② What is Power BI?

Power BI is a suite of business analytics tools by **Microsoft** that helps you connect to data, visualize it and share insights across your organization.



③ Why Organizations Use Power BI?

- ✓ Turns raw data into actionable insights
- ✓ Improves decision making
- ✓ Interactive & real-time dashboards
- ✓ Increases operational efficiency
- ✓ Easy to use (Drag & Drop UI)
- ✓ Enables data-driven culture
- ✓ Connects to multiple data sources
- ✓ Secure & enterprise ready
- ✓ Cost-effective & scalable
- ✓ Seamless integration with Microsoft ecosystem



④ Key Features

- Connect** - Connect to hundreds of data sources.
- Model** - Build a strong data model with relationships.
- Visualize** - Create interactive & beautiful reports.
- Share** - Share insights with your team securely.
- Refresh** - Keep data up-to-date with scheduled refresh.

⑤ Real-world Applications

- Sales Analysis** - Track sales performance, trends & targets.
- Finance Reporting** - P&L, Budget vs Actuals, Cash flow reports.
- Supply Chain** - Inventory, demand forecasting, supplier performance.
- HR Analytics** - Employee performance, attrition, hiring trends.
- Marketing Analysis** - Campaign performance, customer insights, ROI.

Power BI Ecosystem

What is it?

Power BI Ecosystem is a suite of Microsoft tools and services that work together to connect, model, visualize and share data insights across your organization.

Key Components



Power BI Desktop - Free Windows application to connect, transform, model data and create reports.



Power BI Service - Cloud service to publish, share, collaborate, and create dashboards & apps.



Power BI Mobile - Mobile apps to view dashboards & reports on the go.

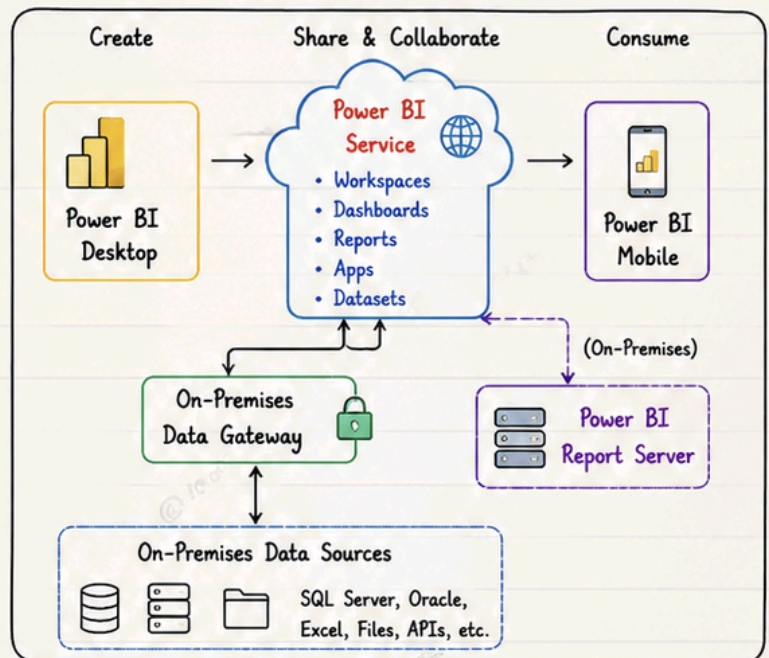


Power BI Report Server - On-premises solution to publish & manage reports within your own environment.



On-Premises Data Gateway - Secure bridge between Power BI Service and on-premises data sources.

How They Work Together



Licensing Overview

Free	Pro	Premium Per User (PPU)	Premium Capacity / Microsoft Fabric
- Power BI Desktop - Publish to My Workspace - Basic sharing (limited) - Not for deployment Best for learning & personal use.	- Full sharing & collaboration - Apps & Workspaces - Scheduled refresh 10 GB dataset storage - Access via mobile Best for individuals & small teams.	- All Pro features - Larger dataset storage (up to 100 GB) - AI features - Paginated reports Best for power users who need more.	- Dedicated capacity - Larger models & faster performance - Supports large-scale enterprise workloads - Includes Microsoft Fabric capabilities Best for enterprise & large organizations.



Key Takeaway: Power BI ecosystem helps you create in Desktop, collaborate in Service, consume on Mobile, extend on-premises with Gateway/Report Server and scale with Premium & Microsoft Fabric.



Interview Tip:

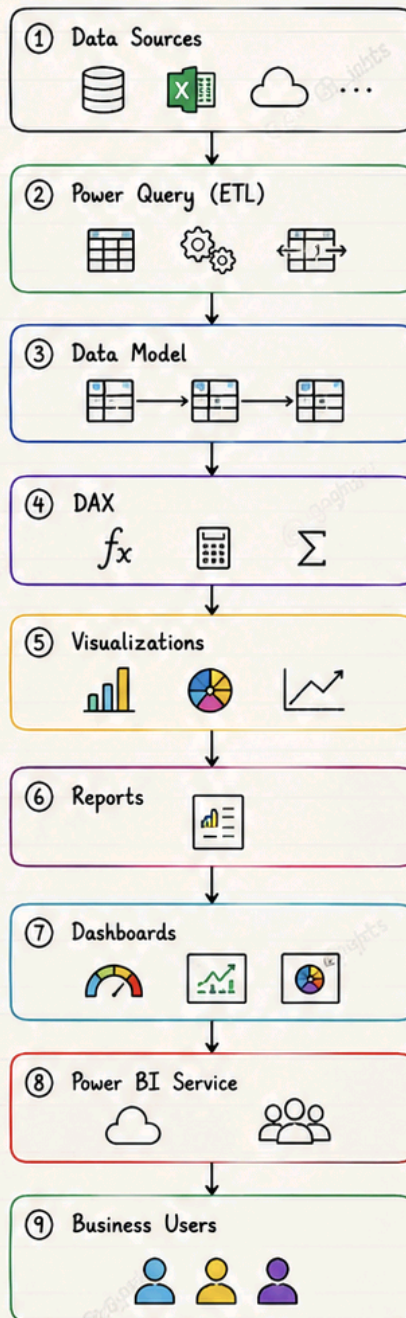
Always choose the right licensing based on team size, data volume, security & performance needs.

Power BI Architecture & Workflow

End-to-End Flow of Power BI from Data to Decisions!

Steps Explained

- ① **Data Sources**
Connect to multiple sources like databases, files, cloud services, APIs, etc.
- ② **Power Query (ETL)**
Extract, clean, transform and load data using Power Query Editor (M Language).
- ③ **Data Model**
Create tables, define relationships and structure data using Star Schema.
- ④ **DAX (Data Analysis Expressions)**
Create measures, calculated columns and tables to perform advanced calculations.
- ⑤ **Visualizations**
Build interactive charts, graphs, cards, KPIs and other visuals to represent insights.
- ⑥ **Reports**
Combine visuals into one or more pages to tell a story with data.
- ⑦ **Dashboards**
Pin key visuals to dashboards for real-time monitoring and quick insights.
- ⑧ **Power BI Service**
Publish reports to the cloud, share, collaborate and schedule refreshes.
- ⑨ **Business Users**
End users access dashboards and reports to make data-driven decisions.



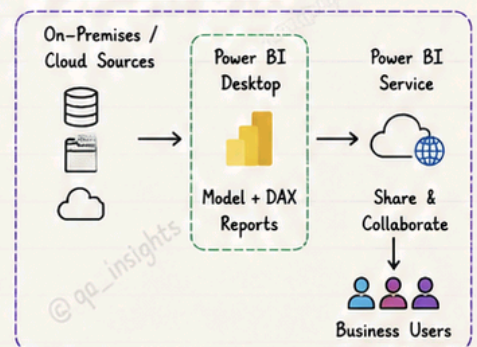
Why This Flow?

- ✓ Transforms raw data into meaningful insights.
- ✓ Ensures clean, reliable and optimized data.
- ✓ Enables interactive and real-time decision making.
- ✓ Supports collaboration and secure sharing.

Key Points 💡

- **Power Query** = Data Preparation
- **Data Model** = Structure & Relationships
- **DAX** = Business Logic & Calculations
- **Reports** = Data Storytelling
- **Dashboards** = At-a-glance Insights
- **Power BI Service** = Share & Collaborate
- **Users** = Data-driven Decisions

Architecture View



Interview Tip

Always remember: Power BI follows a clear flow - Data → Model → Calculate → Visualize → Share → Decide. Mastering each step helps you build powerful and scalable BI solutions!

Connecting Data Sources

Popular Data Sources



Excel - .xlsx, .xls files from local or OneDrive/SharePoint.



SQL Server - Relational data from on-premises or cloud SQL.



Oracle - Data from Oracle Databases.



PostgreSQL - Open-source relational database.



MySQL - Open-source relational database.



CSV - Comma separated values files.



JSON - JavaScript Object Notation files.



SharePoint - Lists, Libraries and SharePoint Online data.



Azure - Azure SQL, Synapse, Blob, Data Lake, Cosmos DB, etc.



REST APIs - Connect to web APIs and pull live data.

How to Connect?

- 1 Home → Get Data → Choose Source
- 2 Authentication (Windows, Database, OAuth, Key, etc.)
- 3 Select Data (Tables, Views, Files, Endpoints)
- 4 Choose Connection Mode (Import / DirectQuery / Live Connection)
- 5 Load data into Power BI Desktop.

Connection Modes Comparison

Feature	Import Mode	DirectQuery Mode	Live Connection
What is it?	Data is imported and stored in Power BI dataset.	Data stays in source. Power BI queries data in real-time.	Connects to Analysis Services model directly.
Performance	Faster (in-memory VertiPaq engine).	Depends on source performance & network.	High (uses existing semantic model).
Data Volume	Best for small to medium datasets.	Best for large datasets.	Depends on model size in Analysis Services.
Data Freshness	Needs Refresh.	Real-time.	Real-time (from model).
Transformations	Full support in Power Query.	Limited transformations.	Handled in source model.
Use Cases	Reports, dashboards, analytics with aggregations.	Real-time operational reporting, large data.	Enterprise semantic model reuse, governed data.
Examples	Excel, CSV, SQL (Import), SharePoint, etc.	SQL Server, Azure SQL, BigQuery, Synapse, etc.	SSAS Tabular Model, Power BI Semantic Model.



Tips

- ✓ Use Import mode for best performance with aggregations.
- ✓ Use DirectQuery when data is huge and changes frequently.
- ✓ Use Live Connection to leverage existing enterprise models.
- ✓ Always enable Query Folding in Power Query for efficiency.
- ✓ Choose the right mode based on data size, latency & refresh needs.



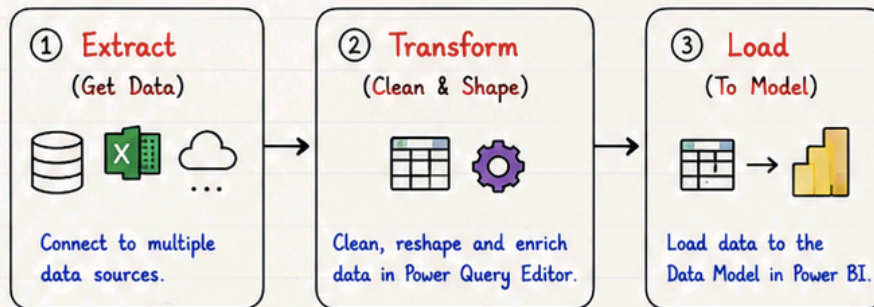
Interview Tip

Know the trade-offs between Import, DirectQuery and Live Connection. Understand when to use each based on business requirements.

Power Query (ETL)

Power Query is the data preparation engine in Power BI.
It helps to connect, transform and load data using **M Language**.

ETL Process



Query Folding

- ✓ Power Query sends your transformations to the data source.
- ✓ Only the required data is retrieved.
- ✓ Improves performance by reducing data load.
- ✓ Works in Import & DirectQuery mode.



Common Transformations

1 Merge Queries Combine rows from two tables based on a common column.	2 Append Queries Combine rows from two or more tables with same structure.	3 Pivot Column Turn unique values in a column into new columns.	4 Unpivot Column Turn multiple columns into attribute-value pairs.
5 Split Column Split a column by delimiter, number of characters, or position.	6 Replace Values Find and replace specific values in a column.	7 Group By Group rows and perform aggregations like Sum, Count, Avg, etc.	8 Remove Duplicates Remove duplicate rows based on one or more columns.



Tips / Best Practices

- ✓ Enable Query Folding for large datasets.
- ✓ Keep transformations at source (DB level) when possible.
- ✓ Use data types early for better performance.
- ✓ Remove unnecessary columns and rows.
- ✓ Reorder steps - Filter early, Aggregate late.



Interview Tip

- Power Query is an ETL tool inside Power BI used for data shaping, cleansing and preparing data before loading into the Data Model.

Key Terms: M Language, Query Folding, Transformations, Steps, Load To, Close & Apply.

Data Modeling in Power BI

Data Modeling is the process of organizing data into tables and relationships to create a logical structure for analysis and reporting.



Tables

A collection of related data organized in rows and columns.



Relationships

Connections between tables using common fields (keys).

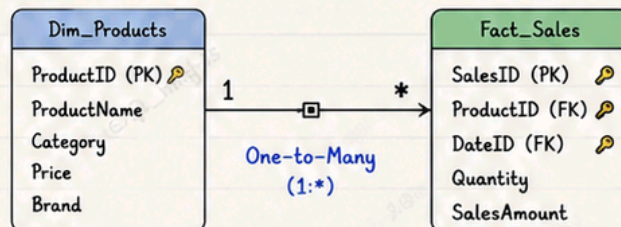


Cardinality

Defines the relationship type between two tables.

Cardinality Types	Description
One-to-One (1:1)	Each row in table A relates to one row in table B.
One-to-Many (1:*)	One row in table A relates to many rows in table B.
Many-to-Many (*:*)	Many rows in table A relate to many rows in table B.

Relationship Components



Active vs Inactive Relationships

- Active relationship is used by default in reports.
- Inactive relationships can be used using DAX (USERRELATIONSHIP).

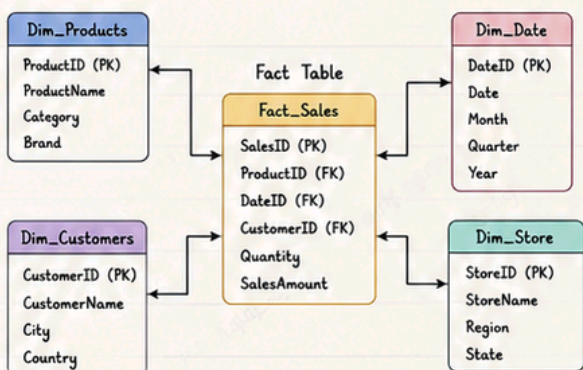


Cross Filter Direction

Controls how filters flow between tables.

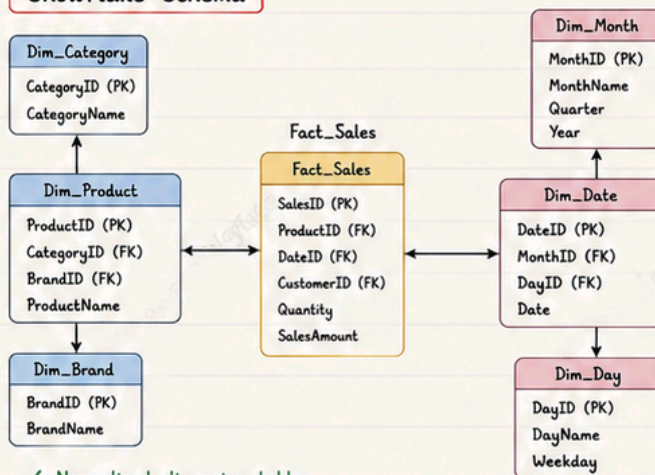
Direction	Description
Single (→)	Filter flows from one table to the other.
Both (↔)	Filter flows in both directions.

Star Schema (Recommended)



- ✓ Simple and easy to understand.
- ✓ Best performance.
- ✓ Ideal for reporting and analytics.

Snowflake Schema



- ✓ Normalized dimension tables.
- ✓ Saves storage space.
- ✓ More complex and slower than Star Schema.



Best Practices

- ✓ Use Star Schema for better performance.
- ✓ Use meaningful table and column names.
- ✓ Keep relationships simple and clear.
- ✓ Hide unnecessary columns and tables.
- ✓ Use surrogate keys instead of natural keys.
- ✓ Avoid bi-directional filter unless required.



Common Mistakes

- ✗ Too many bi-directional relationships.
- ✗ Joining fact tables directly.
- ✗ Using calculated columns unnecessarily.
- ✗ Ignoring data types and cardinality.



Interview Tip

Star Schema is preferred in most projects. Understand when to use Inactive Relationships and how Cross Filter Direction affects your reports.

DAX Fundamentals

What is DAX?

DAX (Data Analysis Expressions) is a formula language used in Power BI, Power Pivot and Analysis Services to perform calculations and retrieve data from the Data Model.

💡 Key Points

- ✓ DAX works on the Data Model (not on raw data).
- ✓ It is case-insensitive.
- ✓ It is similar to Excel formulas but more powerful.
- ✓ Used for creating Measures, Calculated Columns and Calculated Tables.

DAX Objects



Measures

Dynamic calculations that change with filters and context. Used in reports & visuals.



Calculated Columns

Computed column added to a table. Calculated row by row and stored in model.



Calculated Tables

Creates a new table using a DAX expression. Stored in the model like physical tables.

Basic DAX Functions

Function	What it Does	Syntax	Example
SUM	Adds all numeric values in a column.	SUM(ColumnName)	Total Sales = SUM(Sales[Amount])
COUNT	Counts the number of rows in a column.	COUNT(ColumnName)	Total Orders = COUNT(Orders[OrderID])
IF	Returns a value if a condition is TRUE, otherwise another.	IF(Condition, TrueValue, FalseValue)	High Sales = IF([Total Sales] > 100000, "High", "Low")
CALCULATE	Changes filter context to evaluate an expression.	CALCULATE(Expression, Filter1, Filter2, ...)	Sales LY = CALCULATE([Total Sales], 'Date'[Year] = 2023)
FILTER	Returns a filtered table based on a condition.	FILTER(Table, Condition)	Filtered Sales = FILTER(Sales, Sales[Amount] > 1000)
RELATED	Gets a value from related table in many-to-one relationship.	RELATED(ColumnName)	Customer Name = RELATED(Customer[Name])

✓ Tips / Best Practices

- ✓ Use Measures for calculations in visuals.
- ✓ Use Calculated Columns for row-level static data.
- ✓ Prefer Measures over Calculated Columns for performance.
- ✓ Understand Row Context vs Filter Context.
- ✓ Use meaningful names for DAX objects.

★ Interview Tip

- DAX calculations depend on the current context.
- Measures change with filters, Calculated Columns do not.
- CALCULATE() is the most powerful DAX function.
- RELATED() works only in many-to-one relationships.



Note: DAX is at the heart of Power BI. Mastering it helps you build powerful and dynamic reports!

Advanced DAX

1. Row Context

Row Context exists when a formula evaluates each row in a table one at a time. It is created by functions like SUMX, FILTER, ADDCOLUMNS, etc.

Sales Table

Product	Amount
A	100
B	200
C	150

For each row, DAX evaluates expressions individually.

2. Filter Context

Filter Context is the set of filters applied in the report (slicers, filters, visuals, etc.). It determines which data is visible during calculation.

Example:

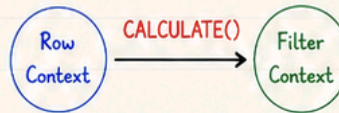
Slicer: Year = 2024

Only data for 2024 is considered.

Year
☐ 2023
☒ 2024

Context Transition

Occurs when DAX converts Row Context into Filter Context. This happens automatically in CALCULATE().



5. DAX Example (Using VAR, CALCULATE, FILTER, DIVIDE)

Total Sales LY % =

VAR SalesLY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))

VAR SalesCY = [Total Sales]

RETURN

DIVIDE(SalesCY - SalesLY, SalesLY, 0)

What it does?

Calculates % change in Total Sales compared to Last Year.

4. Important DAX Functions

SUMX	Iterator function. Adds expression for each row. Ex: SUMX(Sales, Sales[Quantity]*Sales[Price])
RANKX	Returns rank of a value in a table. Ex: RANKX(ALL(Products), [Total Sales])
ALL	Removes all filters from a table or column. Ex: CALCULATE([Total Sales], ALL(Products))
ALLEXCEPT	Removes all filters except specified columns. Ex: ALLEXCEPT(Sales, Sales[Category])
VALUES	Returns distinct values from a column. Ex: VALUES(Products[Category])
SWITCH	Evaluates an expression against multiple conditions. Ex: SWITCH(TRUE(), [Sales] > 1000, "High", "Medium")
DIVIDE	Safe division (handles divide by zero). Ex: DIVIDE([Profit], [Sales], 0)
VAR	Stores a variable to improve readability and performance. Ex: VAR x = [Total Sales]
RETURN	Returns the result of the expression after VAR.

6. Time Intelligence Basics

- **SAMEPERIODLASTYEAR()** - Compares data with same period last year.
- **PREVIOUSYEAR()** - Returns previous year.
- **DATESYTD()** - Returns year to date data.
- **TOTALYTD()** - Calculates YTD total.
- **DATESMTD()** - Returns month to date data.
- **TOTALMTD()** - Calculates MTD total.

Example:

Sales YTD =

TOTALYTD(

[Total Sales],

'Date'[Date]

)

Calculates sales from start of the year up to current date.

★ Interview Tip

- Understand Row vs Filter Context clearly.
- CALCULATE() is the key to DAX.
- Iterator Functions iterate row by row.
- Use VAR for clean, readable & fast expressions.
- Time Intelligence functions work only with proper Date table.

Best Practices

- ✓ Use meaningful names for Measures.
- ✓ Avoid complex calculated columns.
- ✓ Keep Date table marked as Date Table.
- ✓ Prefer Measures over Calculated Columns.
- ✓ Use DAX Studio to test & optimize DAX.
- ✓ Use variables (VAR) to improve performance.

Data Visualization in Power BI

Choosing the right visual is key to building clear, insightful and impactful reports & dashboards.



Visual	Best Use Case	Visual	Best Use Case
 1. Column Chart	Compare values across categories.	 9. Waterfall Chart	Show cumulative effect of positive & negative values.
 2. Bar Chart	Compare values across categories (good for long category names).	 10. Funnel Chart	Show stages in a process and conversion.
 3. Line Chart	Show trend over time (continuous data).	 11. Gauge	Show progress towards a goal.
 4. Area Chart	Show volume / magnitude over time.	 12. KPI	Track performance against a target.
 5. Pie Chart	Show contribution of parts to a whole (few categories).	 13. Maps	Show data by geographic location.
 6. Donut Chart	Similar to Pie with space in the center.	 14. Matrix	Show detailed data in rows & columns with summaries.
 7. Scatter Plot	Show relationship / correlation between two measures.	 15. Cards	Display single KPI or important value.
 8. Treemap	Show hierarchy and part-to-whole comparison.		



Tips / Best Practices

- ✓ Use the simplest visual that conveys the message.
- ✓ Avoid Pie/Donut for more than 5-6 categories.
- ✓ Use consistent colors & themes.
- ✓ Add clear titles, labels & tooltips.
- ✓ Always focus on audience and business goal.



Interview Tip

- Always ask: What story do I want to tell with this data?
- Choose visuals based on the data type and the insight you want to deliver.
- Less is more - avoid clutter!



Common Mistakes

- ✗ Using wrong chart for the data.
- ✗ Too many visuals on one page.
- ✗ Not using proper labels and titles.
- ✗ Using 3D visuals unnecessarily.
- ✗ Ignoring mobile layout and readability.

Dashboard Design in Power BI

A well-designed dashboard converts data into insights instantly.

Focus on clarity, simplicity and user experience. ★

1 UX Principles



- Know your audience & their goals.
- Show the right KPIs, not all data.
- Keep it simple, consistent & intuitive.
- Follow visual hierarchy (important info first).
- Reduce clutter, increase clarity.

2 Color Psychology



- Blue - Trust, professional (best for corporate dashboards)
- Green - Growth, positive, success
- Orange - Energy, attention
- Red - Alerts, issues, critical values
- Grey - Neutral, balance, background



Tip: Use limited colors (3-5) and be consistent.

3 Layout

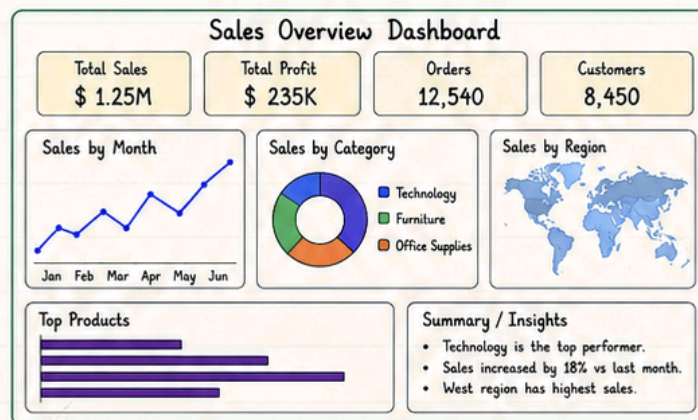


- Use grid layout for alignment.
- Place KPIs at the top.
- Charts in the middle.
- Details at the bottom.
- Follow F-pattern reading (Top → Left → Right → Down).

4 White Space



- White space improves readability.
- Don't crowd visuals.
- Keep breathing room between visuals.
- Helps users focus on what matters.



5 Navigation



- Make navigation simple.
- Use clear page names.
- Provide easy movement between pages.
- Keep important pages within 2-3 clicks.

6 Bookmarks



- Save a report state (filters, selections, visual states).
- Used for scenarios like "Top 10", "View by Region".
- Enable users to switch views quickly.

7 Buttons



- Use buttons for navigation and actions.
- Types: Page Navigation, Bookmark, URL, Drillthrough.
- Keep size consistent and label clearly.

8 Tooltips



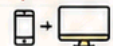
- Provide additional information on hover.
- Use report page tooltips for rich and interactive details.
- Keep tooltips clean and relevant.

9 Drillthrough



- Allows users to go from summary → details.
- Add Drillthrough fields in the drillthrough pane.
- Great for exploring data without cluttering the main page.

10 Responsive Design



- Design for all devices (Desktop, Tablet, Mobile).
- Use relative sizing instead of fixed sizes.
- Test in different viewports.
- Provide a great experience everywhere.



Best Practices

- ✓ Show only what matters.
- ✓ Use consistent colors & fonts.
- ✓ Maintain alignment & grid.
- ✓ Use meaningful titles & labels.
- ✓ Test with real users & iterate.



Interview Tip

Always explain the WHY behind your design choices. Focus on business goals, users, clarity, and actionable insights. Good dashboards drive better decisions!



Common Mistakes

- ✗ Using too many colors & visuals.
- ✗ Showing unnecessary data.
- ✗ Poor alignment & inconsistent layout.
- ✗ Small fonts and low contrast.
- ✗ Ignoring mobile responsiveness.

Filters & User Interaction in Power BI

Filters and interactions help users explore data, focus on important insights and navigate reports easily.



1. Filters in Power BI



Visual Filter Applies to a specific visual. Affects only that visual.



Page Filter Applies to all visuals on the current page.



Report Filter Applies to all pages in the entire report.

2. Slicers



- Visual elements that allow users to filter data interactively.
- Supports single select, multi-select and search.
- Can be synced across pages.



Tip:

Use Slicers for a user-friendly and dynamic report experience.

3. Drill Actions & Navigation



Drill Down
Move from higher level data to lower level detail.

Example:

Year → Quarter → Month → Day



Drill Up
Move from lower level data to higher level summary.

Example:

Day → Month → Quarter → Year

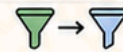


Drillthrough
Navigate from summary page to a detailed target page.

Use Case:

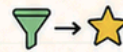
Click on a country → open Country Details page.

4. Interactions Between Visuals



Cross Filtering

- Selecting data in one visual filters other related visuals.
- Helps focus on relevant data.



Cross Highlighting

- Highlights related data in other visuals without filtering.
- Useful for comparison.



Sync Slicers

- Synchronize slicers across multiple pages.
- Keeps the same filter selection on other pages.
- Saves time and improves user experience.

How Filters & Interactions Work?

User Action

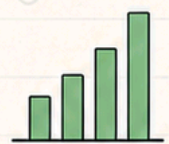
Slicer Selection
Click on Visual
Drill / Navigate

Power BI Engine

Apply Filters
Update Context
Refresh Visuals

Result

Relevant Data
Updated Visuals
Better Insights



Best Practices

- ✓ Use appropriate filter level.
- ✓ Prefer Slicers over advanced filters.
- ✓ Keep filter pane clean and organized.
- ✓ Use cross highlighting for comparisons.
- ✓ Sync slicers for better navigation.



Interview Tip

- Understand the difference between filters and interactions.
- Know the order of filter precedence: Visual > Page > Report.
- Be clear on Drillthrough vs Drill Down.



Common Mistakes

- ✗ Applying too many filters at multiple levels.
- ✗ Using Drillthrough without context.
- ✗ Not syncing slicers on important pages.
- ✗ Confusing Cross Filter with Cross Highlight.

Power BI Service

Power BI Service is the cloud platform (SaaS) where you publish, manage, share and collaborate on your BI assets.



1. Key Components



Publish

Upload reports from Power BI Desktop to Power BI Service.



Workspace

A container to store and manage datasets, reports and dashboards.



Dataset

The semantic model containing data and relationships.



Dataflow

Cloud-based Power Query for clean, reusable data across reports.



Dashboard

Single page view of key metrics and visuals for monitoring KPIs.



Reports

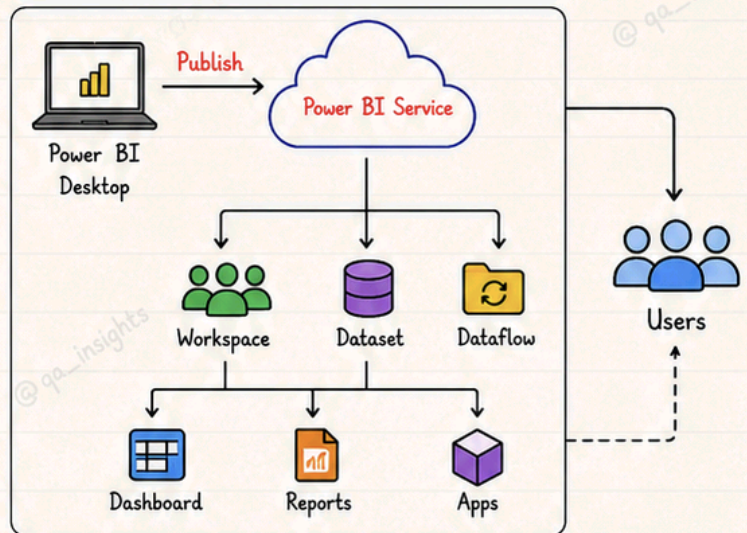
Interactive, multi-page reports with rich visual analytics.



Apps

Package and distribute content to users in your organization.

2. How It Works?



Benefits

- ✓ Centralized cloud platform for BI collaboration.
- ✓ Secure access and sharing control.
- ✓ Real-time insights on any device.
- ✓ Scalable and enterprise ready.

3. Sharing & Collaboration

- **Sharing:** Share reports/dashboards with specific users or security groups.
- **Role-based access:** Admin, Member, Contributor, Viewer.
- **Comments & @mentions:** Discuss insights directly on visuals.
- **Co-authoring:** Multiple users can work together in the same workspace.

4. Scheduled Refresh

- Keep your data up-to-date automatically.
- Configure refresh in **Dataset** settings.
- Requires **Gateway** for on-premises data sources.
- Supports **Incremental Refresh** for large datasets.



Best Practices

- ✓ Use meaningful workspace and item names.
- ✓ Organize content using folders and apps.
- ✓ Use Dataflows for certified data.
- ✓ Enable Incremental Refresh for performance.
- ✓ Monitor refresh failures and usage.



Interview Tip

- Workspace is like a container.
- Dataset is the source of truth.
- Dashboard is 1-page summary.
- Reports are detailed & interactive.
- Apps help in easy distribution.



Common Mistakes

- ✗ Publishing sensitive data without RLS.
- ✗ Sharing with "Anyone" by mistake.
- ✗ Too many datasets with same data.
- ✗ Not configuring refresh schedules.
- ✗ Ignoring Gateway for on-prem data.

Security in Power BI



Power BI provides multiple security layers to protect data, control access and ensure compliance.



① Row-Level Security (RLS)

- Restricts rows of data visible to users.
- Defined in the data model using DAX filter expressions.
- Ensures users see only their own relevant data.



② Object-Level Security (OLS)

- Hides specific tables, columns or measures from certain users.
- Helps protect sensitive fields within the dataset.



③ Workspace Roles

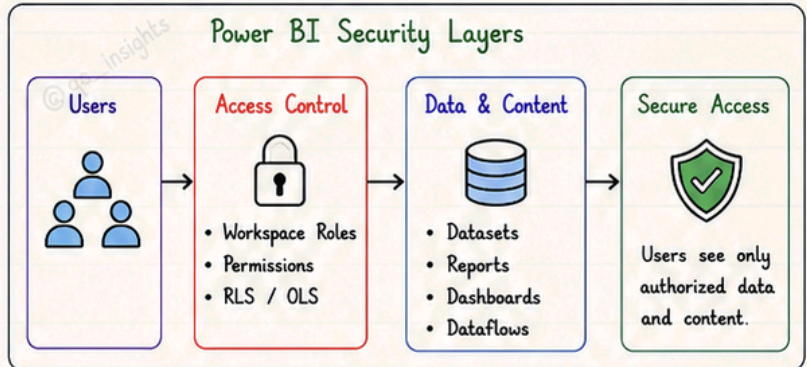
- Admin – Full control over workspace.
- Member – Create content and share.
- Contributor – Create & edit content.
- Viewer – View content only.



④ Permissions

- Control access at multiple levels: Workspace, App, Dashboard, Report and Dataset.
- Set permissions while sharing or using Apps.

Power BI Security Layers



⑤ Data Privacy

- Classify and combine data from different sources.
- Levels: None, Organizational, Private, Confidential.
- Prevents data leakage across sources (via Privacy Firewall).
- Helps control data movement and sharing.

Privacy Levels

None < Organizational
< Private < Confidential

⑥ Sensitivity Labels

- Apply labels like Public, Internal, Confidential, Highly Confidential.
- Labels can encrypt data and restrict sharing.
- Integrated with Microsoft Purview Information Protection.

- Public
- Internal
- Confidential
- Highly Confidential

⑦ Gateway Security

- On-premises Data Gateway securely connects on-prem data to Power BI Service.
- Supports Windows Authentication, Kerberos and Service Principal.
- Encrypts data in transit and stores credentials securely.
- Gateway Admins control data source access and permissions.

RLS vs OLS

Feature	RLS (Row-Level Security)	OLS (Object-Level Security)
What it controls	Rows of data	Tables, Columns, Measures
Where applied	Data Model (Roles)	Data Model (Permissions)
Uses DAX	Yes	No
Dynamic per user	Yes	No
Best for	User-specific data	Hide sensitive fields
Example	Sales rep sees only their region data	Hide 'Salary' column from all users

💡 Best Practices

- ✓ Always implement RLS in the data model, not in reports.
- ✓ Use least privilege access.
- ✓ Test RLS with "View as Role".
- ✓ Apply Sensitivity Labels for important data.
- ✓ Secure your Gateway and refresh credentials.

★ Interview Tip

- RLS filters rows dynamically based on the logged-in user.
- OLS hides objects completely, users won't even know they exist.
- Workspace roles + permissions control access to content.
- Use Data Privacy levels to prevent data leakage.

Performance Optimization in Power BI

Optimizing your Power BI model and reports ensures faster load times, better user experience and reduced resource usage. ★

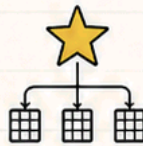


① Query Folding



- Enables Power Query to send transformations to the source.
- Reduces data transfer and improves performance.

② Star Schema



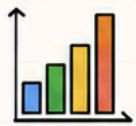
- Use Star Schema with fact and dimension tables.
- Improves DAX performance and simplifies relationships.

③ Optimize Relationships



- Avoid many-to-many relationships.
- Use single direction filter whenever possible.
- Remove unnecessary relationships.

④ Reduce Visuals



- Avoid too many visuals on a single page.
- Use meaningful visuals that add business value.

⑤ Incremental Refresh



- Refresh only new/changed data instead of full dataset.
- Greatly improves refresh time for large models.

⑥ Performance Analyzer



- Identify visuals and DAX queries causing slow performance.
- Use it to optimize report rendering time.

⑦ Optimize DAX



- Use variables (VAR) to store intermediate results.
- Avoid complex calculations in row context.
- Use efficient functions.

⑧ Remove Unused Columns



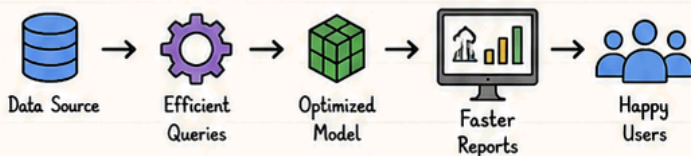
- Remove unnecessary columns from Power Query.
- Reduces model size and memory usage.

⑨ Aggregations



- Create Aggregation Tables for large datasets.
- Redirects queries to smaller summary tables for speed.

How Optimization Improves Performance



Less Data Transfer → Lower Memory Usage → Faster Performance



Best Practices

- ✓ Design your model before building reports.
- ✓ Always prefer Import Mode for performance.
- ✓ Keep your data model clean and lean.
- ✓ Use proper data types and avoid text columns.
- ✓ Monitor performance regularly.



Common Mistakes

- ✗ Importing unnecessary columns and tables.
- ✗ Creating many-to-many relationships.
- ✗ Using complex DAX in calculated columns.
- ✗ Adding too many visuals on one page.
- ✗ Ignoring Performance Analyzer insights.



Interview Tip

- Explain how you improved report performance in a real project.
- Mention tools like Performance Analyzer, DAX Studio, and DAX optimization techniques.

DAX Example (Use of VAR)

```

Total Sales (Optimized) =
VAR TotalQty = SUM(Sales[Quantity])
VAR TotalAmt = SUM(Sales[Amount])
RETURN
IF(TotalQty > 0, TotalAmt, 0)
  
```

Using VAR improves readability and performance.

Real-Time Business Project Workflow

A complete end-to-end Power BI project lifecycle in real-world industry. ★

Goal:



Transform raw data into meaningful insights and deliver value to business users.

Industry Example



Sales Performance Dashboard for a Retail Company

Tools Used

- Power BI Desktop
- Power Query
- DAX
- Power BI Service
- Gateway (if needed)
- Excel / SQL / APIs
- Data Source



① **Business Requirement** • Understand business goals, KPIs and success metrics.



② **Data Collection** • Collect data from multiple sources (DB, Excel, APIs, etc.).



③ **Power Query (Extract)** • Connect & import data using Power Query Editor.



④ **Data Cleaning** • Remove duplicates, handle nulls, standardize and shape data.



⑤ **Data Model** • Create relationships, define Star Schema, optimize model.



⑥ **DAX (Calculations)** • Create Measures, Calculated Columns and advanced business logic.



⑦ **Visualization** • Build interactive visuals, apply slicers, tooltips and formatting.



⑧ **Testing** • Validate data accuracy, visuals, filters, DAX and performance.



⑨ **Publish** • Publish report to Power BI Service in appropriate workspace.



⑩ **User Acceptance Testing** • Business users validate reports and provide feedback.



⑪ **Production** • Go live! Enable sharing, refresh schedule and monitor usage.

Key Outcomes 🏆

- ✓ Actionable insights
- ✓ Better decisions
- ✓ Faster reporting
- ✓ Improved efficiency
- ✓ Business growth

💡 Best Practices

- ✓ Understand business before building.
- ✓ Use Star Schema for better performance.
- ✓ Keep data clean and consistent.
- ✓ Write efficient DAX.
- ✓ Test at every stage.
- ✓ Monitor & optimize continuously.

★ Interview Tip

Always mention the end-to-end workflow when asked about Power BI projects. It shows real-world experience!

Power BI Interview Questions



Top 20 most important Power BI interview questions with short, crisp answers.

① What is the difference between Import and DirectQuery?



- Import stores data in memory for fast analysis. DirectQuery fetches data in real time from source.

② What is the difference between DAX and M Language?



- DAX is used for data analysis & calculations. M Language is used in Power Query for data transformation.

③ What is the difference between Measure and Calculated Column?



- Measure is calculated on the fly and uses less memory. Calculated Column is stored in the model and increases memory.

④ What is Row-Level Security (RLS)?



- RLS restricts data visibility at row level based on user roles.

⑤ What is Star Schema?



- A data modeling technique with one fact table connected to multiple dimension tables.

⑥ What is Query Folding?



- It pushes transformations to the data source to improve performance.

⑦ What is a Gateway in Power BI?



- Gateway acts as a bridge between on-premises data sources and Power BI Service.

⑧ What is the difference between Dataset and Report?



- Dataset is the data model. Report is the visual representation built on top of the dataset.

⑨ What is the difference between Dashboard and Report?



- Report is interactive and ad-hoc. Dashboard is single page summary of key visuals (tiles).

⑩ What is Incremental Refresh?



- It refreshes only new or changed data partitions, improving refresh time for large datasets.

⑪ What does CALCULATE() function do in DAX?



- It changes the filter context to evaluate an expression.

⑫ What is Filter Context?



- Filter Context is the set of filters applied to tables/columns that affect DAX calculations.

⑬ What is Context Transition?



- Converts Row Context to Filter Context (happens automatically in CALCULATE()).

⑭ What is a Dataflow?



- A dataflow prepares and stores data in the cloud for reuse across datasets and reports.

⑮ What are Workspaces?



- Workspaces are containers to organize and collaborate on Power BI content.

⑯ What is the difference between Cross Filter and Cross Highlight?



- Cross Filter filters data in other visuals. Cross Highlight highlights related data only.

⑰ How can you optimize a slow Power BI report?



- Optimize model, reduce visuals, use DAX best practices, enable Aggregations, use Performance Analyzer.

⑱ What are Bookmarks?



- Bookmarks save the state of a report page (filters, slicers, visuals) for user navigation.

⑲ What is Drillthrough?



- Drillthrough navigates from a summary visual to a detailed target page.

⑳ What is the best practice for data modeling?



- Use Star Schema, proper data types, avoid bi-directional relationships, keep model clean.



Best Practices

- ✓ Model for performance, not just functionality.
- ✓ Use measures over calculated columns.
- ✓ Keep DAX simple and reusable.
- ✓ Test with real data & optimize.



Interview Tip

- Explain with examples.
- Mention real-world use cases.
- Be clear, structured and concise.
- Practice DAX and scenarios.



Common Mistakes

- ✗ Using bi-directional relationships everywhere.
- ✗ Overusing calculated columns.
- ✗ Ignoring Query Folding.
- ✗ Not testing report performance.

Power BI Best Practices & Common Mistakes

Follow best practices to build scalable, maintainable and high-performance Power BI solutions and avoid common mistakes.



✓ Best Practices

① Naming Conventions



- Use clear, consistent and meaningful names for tables, columns, measures.
- Use prefixes: tbl_, dim_, fact_, msr_ for measures.

② Folder Organization



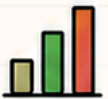
- Organize reports, datasets and dashboards in logical folders.
- Keep workspace clean and easy to navigate.

③ Reusable Measures



- Create reusable measures for calculations.
- Avoid hardcoding values and use variables where possible.

④ Model Optimization



- Use proper data types and remove unused columns.
- Use Star Schema and keep model small and efficient.

⑤ Consistent Themes



- Use consistent colors, fonts and layouts.
- Follow company branding and design standards.

⑥ Documentation



- Document data sources, transformations, measures and business logic.
- Helps in maintenance and knowledge sharing.



Additional Tips

- ✓ Always validate data accuracy before publishing.
- ✓ Use Power Query for data preparation, not in DAX.
- ✓ Monitor performance using Performance Analyzer.
- ✓ Review and optimize data model regularly.

✗ Common Mistakes

① Flat Tables



- Using flat tables without relationships leads to redundant data and wrong results.
- Always use Star Schema.

② Too Many Visuals



- Overloaded reports confuse users and slow down performance.
- Keep visuals minimal and meaningful.

③ Duplicate Calculations



- Creating same logic in multiple measures or columns increases model size.
- Use reusable measures and variables.

④ Poor Relationships



- Incorrect or many-to-many relationships lead to wrong results and poor performance.
- Use proper cardinality and single direction.

⑤ Ignoring Performance



- Ignoring DAX optimization, model size and refresh times.
- Always test and monitor performance.

⑥ Excessive Custom Visuals



- Too many custom visuals can increase load time and maintenance effort.
- Use only when built-in visuals are insufficient.



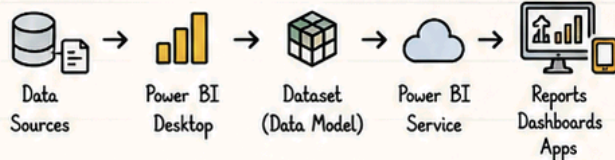
Interview Tip

- How do you optimize a slow report?
- What is the ideal data model in Power BI?
- How do you reduce model size?
- What are the key things to avoid in Power BI projects?

Power BI Cheat Sheet & Roadmap

Quick revision of key concepts, workflows and roadmaps
to become a Power BI Expert! ★

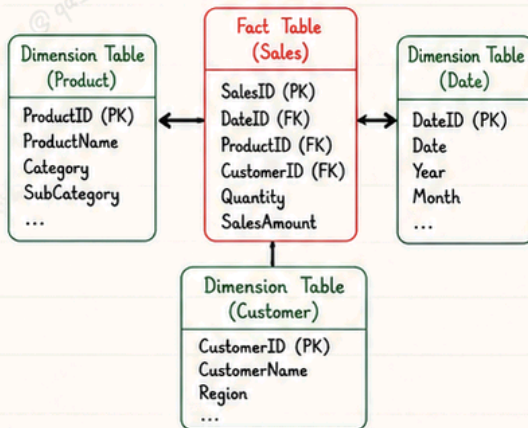
1) Architecture Overview



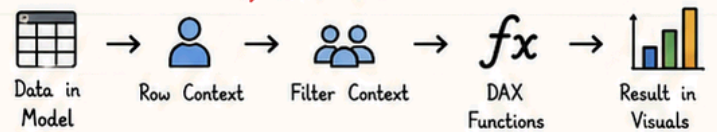
2) ETL Flow (Power Query)



3) Data Modeling Essentials



4) DAX Flow



5) Visualization Selection Guide

	Column Chart	Compare values across categories.		Treemap	Show part-to-whole in hierarchical rectangles.
	Bar Chart	Compare values across categories (long labels).		Waterfall	Show cumulative effect (positive/negative).
	Line Chart	Show trends over time.		Funnel	Show stages in a process (drop-off).
	Area Chart	Show volume over time.		Gauge	Show progress towards a goal.
	Pie Chart	Show part-to-whole (only few categories).		KPI	Display key metrics vs targets.
	Donut Chart	Part-to-whole (better than Pie).		Maps	Show data by geography.
	Scatter Plot	Find correlation between two measures.		Matrix	Display multi-dimensional data with totals.

6) Deployment Steps

1. Develop report in Power BI Desktop.
2. Publish to Power BI Service.
3. Configure Data Source & Gateway (if needed).
4. Set up Scheduled Refresh.
5. Create Dashboard from report.
6. Share via Workspace / Apps.
7. Set permissions (RLS, roles, access).
8. Monitor & maintain.



Performance Checklist

- ✓ Use Query Folding in Power Query.
- ✓ Follow Star Schema.
- ✓ Remove unused columns & tables.
- ✓ Use proper data types.
- ✓ Create aggregations for large datasets.
- ✓ Use Incremental Refresh.
- ✓ Optimize DAX (avoid complex calculations).
- ✓ Use Performance Analyzer.



Interview Revision Box

- Import vs DirectQuery
- DAX vs M Language
- Measure vs Calculated Column
- RLS vs OLS
- Star Schema vs Snowflake Schema
- Context (Row vs Filter vs Query)
- CALCULATE(), RELATED(), ALL(), VALUES()
- Gateway & Refresh Types
- Dataset vs Report vs Dashboard
- Best practices & performance tips

9) Power BI Learning Roadmap

